

Module 1: Core API & AI Engine (Flask + Gemini)

This module acts as the orchestration layer and analysis engine, handling incoming requests and interacting with the LLM.

- **Prompt Engineering & Extraction:** Instruct the Gemini API to parse job offers and return a structured JSON object containing identified red flags, severity levels, and evidence.
- **Deterministic Scoring:** Calculate the 0–100% Scam Threat Index by applying predefined weights (e.g., +30 for upfront payment, +15 for urgency) to the AI-extracted signals.
- **API Surface:** Maintain the `/api/analyze` (POST) endpoint for processing text or URLs, and `/api/health` (GET) for service checks.

Module 2: Domain Intelligence

This module isolates the external domain logic required for URL analysis.

- **Domain Age Lookup:** Extract the domain from a submitted URL and query WHOIS records to determine the registration age.
- **Risk Integration:** Pass the domain age back to the scoring engine to increase the threat index for newly registered domains (e.g., < 30 days old).
- **Graceful Degradation:** Ensure the application falls back to text-only analysis if the domain lookup fails or errors out.

Module 3: Bento UI Frontend (HTML/CSS)

This module focuses purely on the visual structure and responsive design outlined in the design document.

- **Grid System:** Construct the 12-column CSS grid layout supporting Desktop, Tablet (6-column), and Mobile (1-column stack) breakpoints.

- **Color & Typography:** Implement the strict Navy Blue (#092634), Blue (#004E72), Orange (#FF6E42), and White (#F9F9F9) color system using CSS variables.
- **Component Architecture:** Build the required visual cards, including the Input toggler, Threat Index meter, individual Signal cards, and AI Explanation blocks with 18–24px corner radiuses.

Module 4: Client Logic & Data Persistence (JS)

This module wires the user interface to the backend API and manages application state.

- **State Management:** Handle UI transitions across Empty, Loading (showing three lightweight steps), Success, Invalid Input, and API Failure states.
- **DOM Manipulation:** Dynamically render the bento grid cards with the JSON payload returned from `/api/analyze` without requiring a page refresh.
- **Database Integration:** Implement the chosen database (Firebase or Supabase) to persist scan metadata and results remotely.

Basic Folder structure

phishing-inspector_promptwars/

- └─ .env
- └─ Pw_requirements.txt
- └─ Pw_app.py
- └─ Pw_services/
- | └─ __init__.py
- | └─ Pw_analyzer.py # LLM signal extraction + deterministic scoring
- | └─ Pw_domain_intel.py # Domain age & WHOIS lookup
- | └─ Pw_database.py # Stubbed ready for Firebase or Supabase integration
- └─ Pw_static/
- | └─ Pw_css/
- | | └─ Pw_style.css # Bento UI styling (Navy, Blue, Orange, White palette)
- | └─ Pw_js/
- | └─ Pw_app.js # Single-page dynamic fetch and DOM rendering
- └─ PW_templates/
- └─ Pw_index.html # 12-column Bento Grid layout